

Jackson Parrack

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Education

Georgia Institute of Technology | Atlanta, GA

Aug. 2026 - Present

Master of Science in Computer Science, Computing Systems

Saint Louis University | Saint Louis, MO

Aug, 2022 - May 2026

Bachelor of Science in Computer Engineering, Minor in Computer Science

GPA: 3.93/4.0

Sogang University | Seoul, South Korea | Exchange Program

Spring 2024

Experience

Mastercard

Hybrid | O'Fallon, MO

Software Engineer I

Aug. 2026 - Present

- Integrated an internal MCP platform across 6 teams, allowing for queries of Oracle databases and company knowledge using natural language and agentic interfaces such as GitHub Copilot.
- Delivered a modernized clearing solution using Java Spring microservices deployed to Kubernetes, improving scalability while maintaining data transformation accuracy at a max throughput of 14M transactions/day.
- Leveraged LLMs to interface with transaction databases to generate insights about transaction data, requiring strict guardrails and harnesses, and contributing to faster feedback cycles.
- Utilized a transaction simulation application in Go to test functionality of systems end-to-end through the full payment lifecycle, improving test accuracy and reducing production defects.
- Contributed to the integration of a modernized settlement solution, including a shift to cloud-native architecture.
- Built Splunk dashboards to monitor services for failures, discrepancies, and latency spikes.

DataServ

Remote | St. Louis, MO

DevOps Intern

May 2025 - Aug. 2025

- Leveraged custom Python automation scripts to migrate 3,000+ SQL stored procedures across 71 client environments into Git, allowing for easier future version control.
 - Redesigned and implemented 17 post-import SQL configurations across 7 clients, replacing time-based jobs with event-driven triggers and validating reliability through full end-to-end testing.
 - Optimized large-scale data ingestion processes, including a lookup redesign for 21 clients and multiple enterprise scheduling upgrades, improving system stability and performance.
 - Delivered project demonstrations to leadership, communicated progress across teams, and collaborated on troubleshooting complex SQL and configuration issues in a fast-paced Agile environment.
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Projects

Rapid Photonic Innovation Device (RAPID) | ECE Capstone Design Award Winner

Aug. 2025 - Apr. 2026

- Reverse-engineered laser diode assemblies, spindle and stepper motors using oscilloscopes and multimeters, and constructed motor and voice-coil driver circuits from component-level datasheets.
 - Implemented motor and voice-coil control logic in VHDL on a Zynq FPGA, and built the PC-to-FPGA communication pipeline in C with a custom UART protocol and input parser.
 - Built a real-time Python GUI for live pattern visualization and manual hardware control, producing a functional laser-lithography system capable of writing photonic patterns onto a spinning disc.
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Skills

- Programming Languages: C, C++, Python, Java, JavaScript, Go, SQL, Assembly, HTML/CSS
- Tools/Platforms: Android, iOS, AWS, GCP, Git, Linux
- Hardware: VHDL, FPGA, ARM & AVR Microcontrollers, Arduino, Raspberry Pi
- Languages: Spanish (Advanced), Korean (Intermediate), Chinese (Beginner)